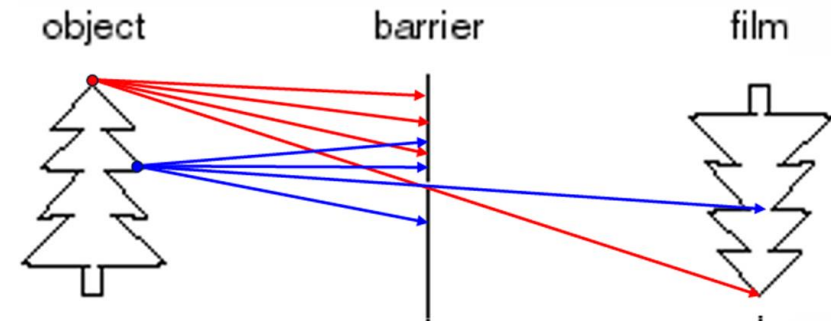
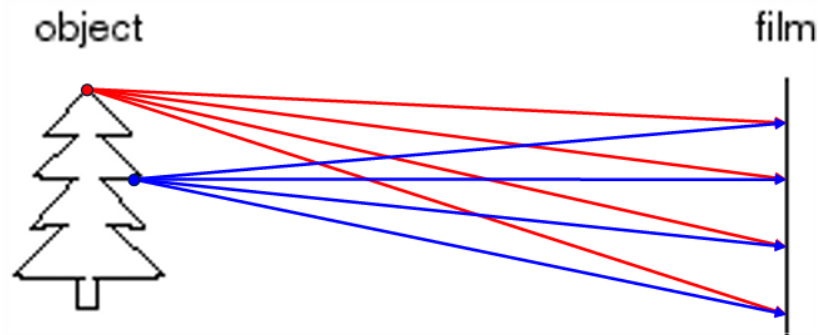


6 - Epipolar Geometry

Projections, Camera Calibration, Epipolar Geometry, Stereo Correspondence

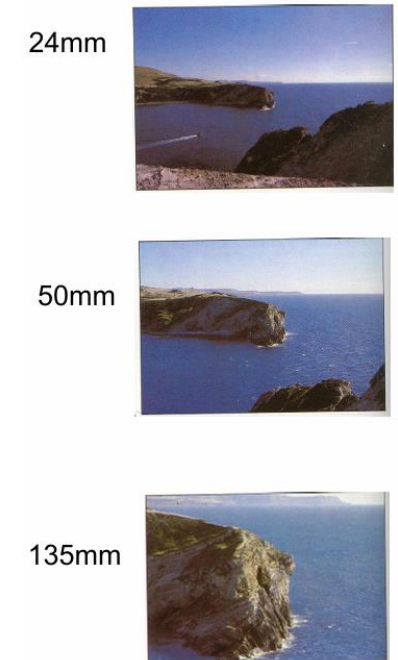
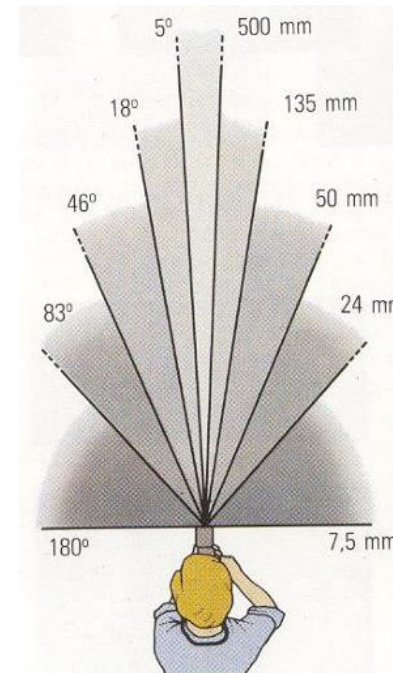
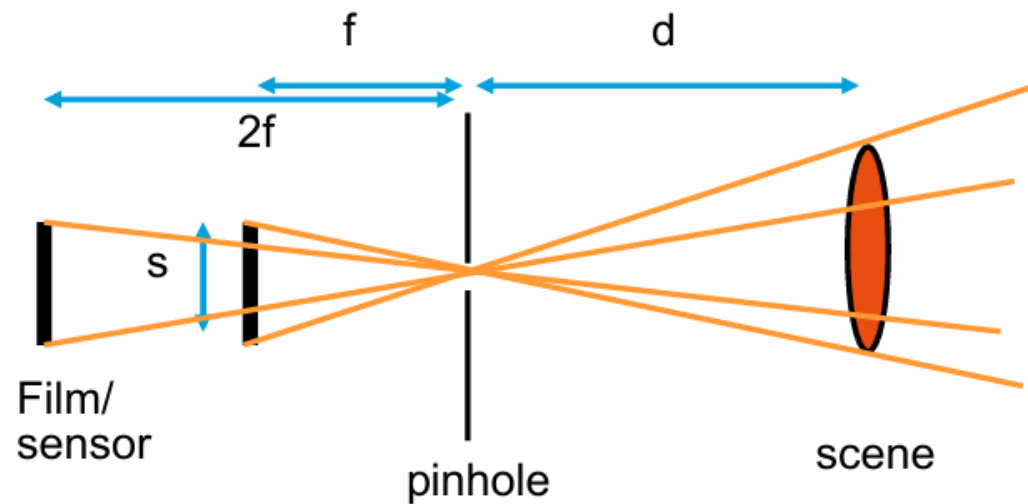
Cameras

- Images can't be captured by placing a piece of film in front of an object; the light will go everywhere and create a blurry image.
- A **pinhole camera** reduces blurring by adding a barrier with a small opening (the **aperture**) in between the object and the film.



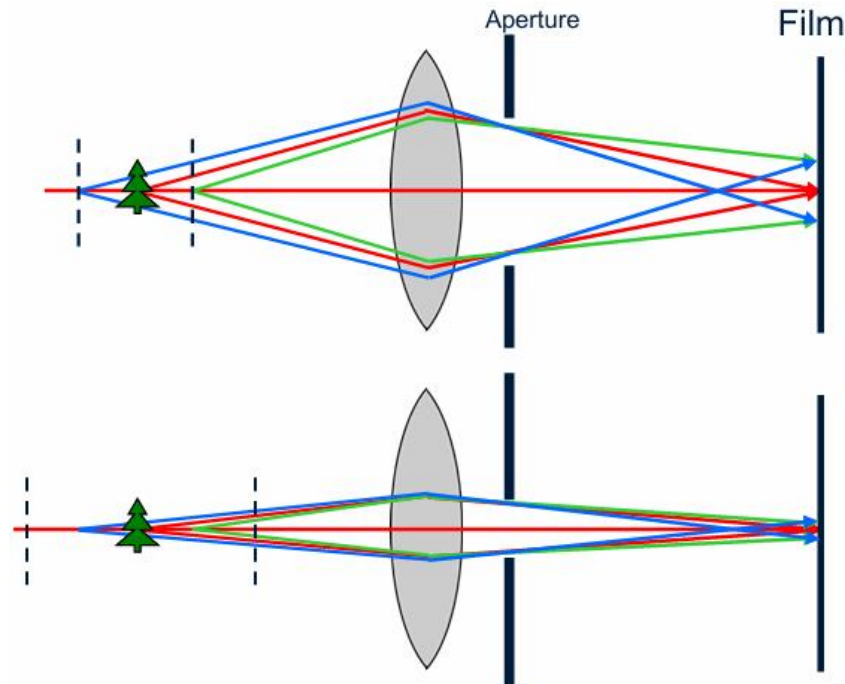
Cameras

- The distance between the film and the barrier is called the **focal length**.
- Increasing the focal length increases the size of the object's projection.



Cameras

- Modern cameras guide light through the aperture using a (series of) lens(es).
- Only points at a certain distance from the lens appear in focus, other points project to a *circle of confusion* on the film, resulting in higher blurriness.
- Smaller aperture sizes increase the range from which an object is in focus.



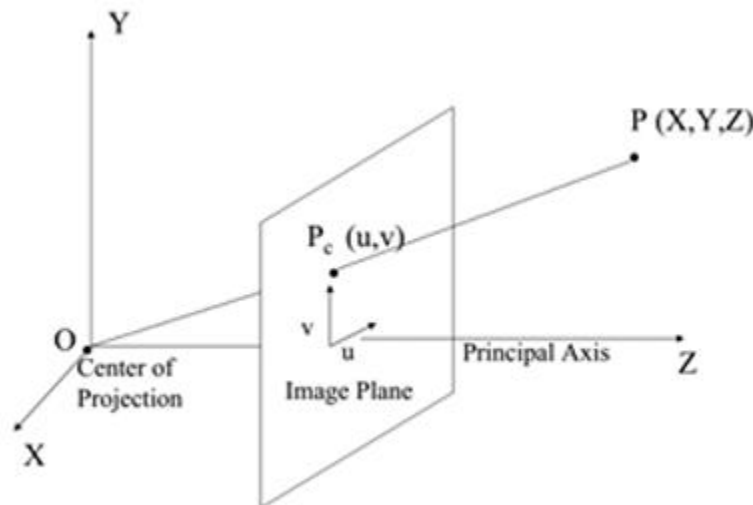
$f/5.6$



$f/32$

Projections

- A 3D point in the real world at $P(X, Y, Z)$ is projected on the camera's *image plane* at coordinate $P_c(u, v)$ (not the same as pixels!).
- The **Center of Projection (COP)** is placed at the origin of the coordinate system, at a distance of f (focal length) from the image plane (**PP**).
- This gives us: $u = \frac{fX}{Z}$, $v = \frac{fY}{Z}$



Projections

- **Perspective Projection:** mimics human vision (objects appear smaller with distance, parallel lines converge, non-linear distortion, ...).
- **Weak Perspective:** a relaxation of *perspective projection* that assumes the depth of an object is small relative to its distance from the camera (objects appear smaller with distance, parallel lines stay parallel, linear distortion, ...).
- **Orthographic Projection:** a relaxation of *weak perspective* that maintains constant scale (all objects appear the same size, parallel lines stay parallel, linear distortion, ...): $\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} X \\ Y \end{bmatrix}$ (**Parallel Projection:** map (X, Y, Z) to (X, Y))

Projections - Homogeneous Coordinates

- To allow affine transformations to be represented as matrix multiplications, n -d points are often represented as **Homogeneous Coordinates** ($n + 1$):

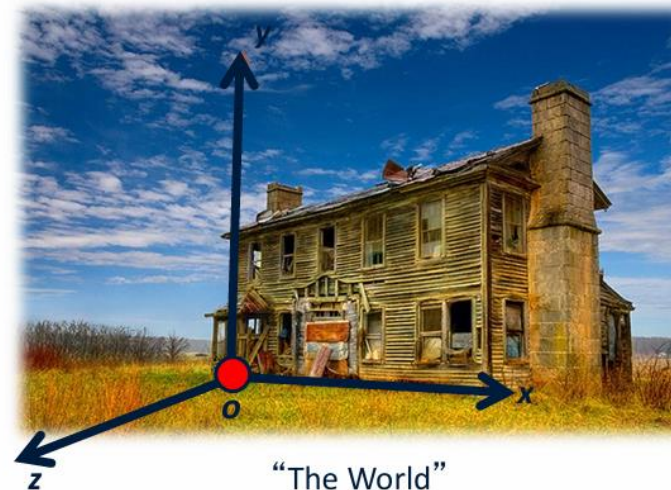
$$\begin{bmatrix} u \\ v \end{bmatrix} \rightarrow \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}$$

- To find image point u, v we can use the following formula:

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$
$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} u/w \\ v/w \end{bmatrix}$$

Camera Calibration

- How to project a point in *world coordinates* (X, Y, Z) in *camera coordinates*?
- To do this, we need to know the camera's position and orientation in world coordinates (**Extrinsics**).
- To transform this into pixels, we need to know the camera's **Intrinsics**.



Camera Calibration - Intrinsics

- Camera intrinsics can be expressed in the following matrix form:

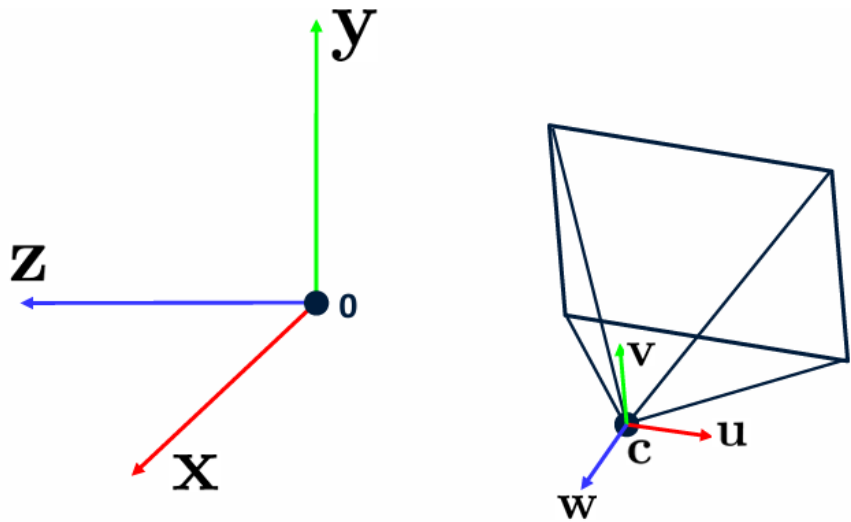
$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} m_u f & 0 & m_u t_u & 0 \\ 0 & m_v f & m_v t_v & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} a_x & 0 & u_0 \\ 0 & a_y & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} P = K \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} P$$

- (t_u, t_v) is the translation of the origin of the image plane to make it coincide with where the Z -axis intersects the image plane.
- (m_u, m_v) is the camera **resolution** (in pixels/meter).
- Matrix K is the **Intrinsic Parameter Matrix**, depending only on the focal length, principal axis location, and resolution (and sometimes a skew parameter s).

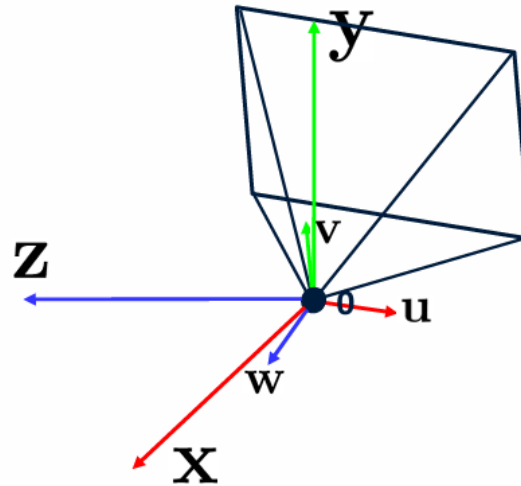
$$K = \begin{bmatrix} a_x & s & u_0 \\ 0 & a_y & v_0 \\ 0 & 0 & 1 \end{bmatrix}$$

Camera Calibration - Extrinsics

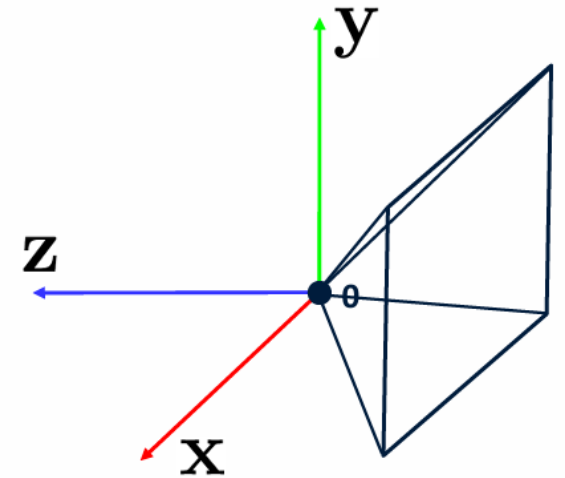
- To get the camera to '*canonical form*' (COP at origin, x-axis pointing right, y-axis pointing up, z-axis pointing backwards), we apply a translation $-c$, followed by a rotation R :
$$P = \begin{bmatrix} R_{3 \times 3} & 0_{3 \times 1} \\ 0_{1 \times 3} & 1 \end{bmatrix} \begin{bmatrix} I_{3 \times 3} & -c_{3 \times 1} \\ 0_{1 \times 3} & 1 \end{bmatrix} P_w$$



Step 1: Translate by $-c$



Step 2: Rotate by R



Camera Calibration

- The cumulative effect of all parameters:

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \underbrace{\begin{bmatrix} a_x & s & u_0 \\ 0 & a_y & v_0 \\ 0 & 0 & 1 \end{bmatrix}}_{\text{intrinsics}} \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{\text{projection}} \underbrace{\begin{bmatrix} R_{3 \times 3} & 0_{3 \times 1} \\ 0_{1 \times 3} & 1 \end{bmatrix}}_{\text{rotation}} \underbrace{\begin{bmatrix} I_{3 \times 3} & -c_{3 \times 1} \\ 0_{1 \times 3} & 1 \end{bmatrix}}_{\text{translation}} P_w = C P_w$$

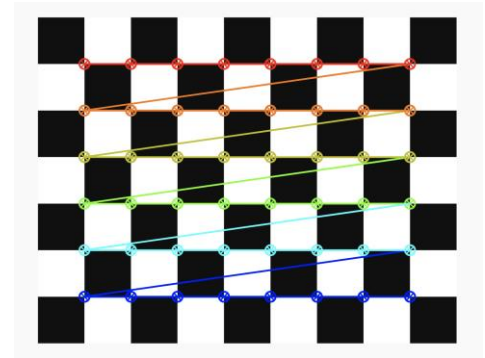
- Where C is the **Camera Matrix**, mapping 3D world coordinates P_w to 2D image pixel coordinates (u, v) :

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = C \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \rightarrow C = \begin{bmatrix} c_{11} & c_{12} & c_{13} & c_{14} \\ c_{21} & c_{22} & c_{23} & c_{24} \\ c_{31} & c_{32} & c_{33} & 1 \end{bmatrix}$$

- C is equivalent to any sC (with $s \neq 0$) because of homogeneous coordinates.

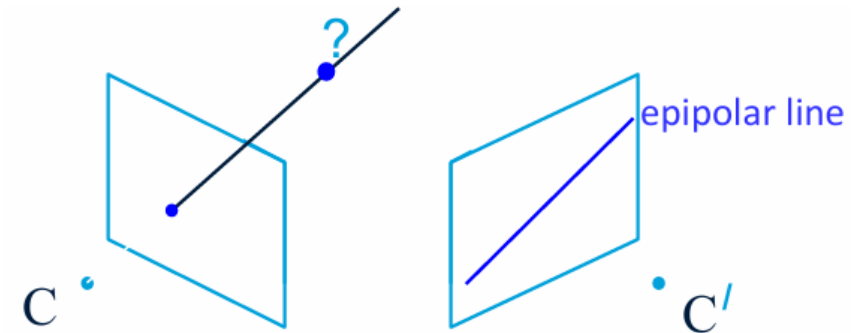
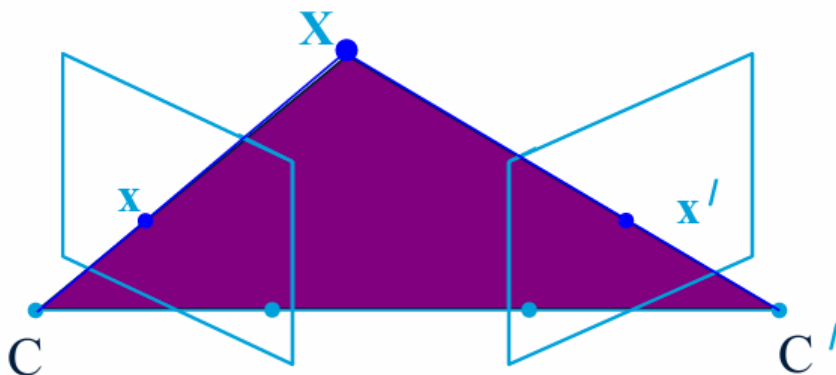
Camera Calibration

- One way to compute $\mathcal{C}$ is to manually establish correspondences between points in 3D world coordinates and pixels in the image:
 - Suppose we know $(x_i, y_i, z_i, 1)$ maps to $(u_i, v_i, 1)$.
 - This gives us the following equations:
 1. $s_i u_i = c_{11}x_i + c_{12}y_i + c_{13}z_i + c_{14}$
 2. $s_i v_i = c_{21}x_i + c_{22}y_i + c_{23}z_i + c_{24}$
 3. $s_i = c_{31}x_i + c_{32}y_i + c_{33}z_i + 1$
 - Through some rewriting we can transform this into a system of linear equations with 11 unknowns and 2 equations.
 - To find an exact solution for $\mathcal{C}$ we need at least 6 pairs of known points (usually found through a checkerboard pattern).
 - If the intrinsic and extrinsic matrices are needed, perform **RQ-decomposition** on the left-most 3x3 submatrix of $\mathcal{C}$.



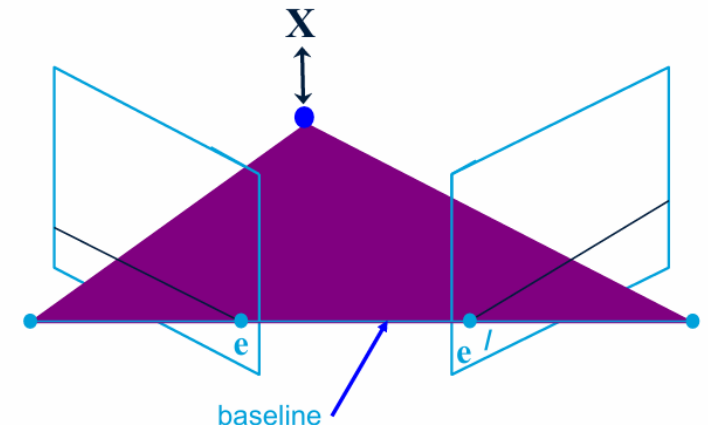
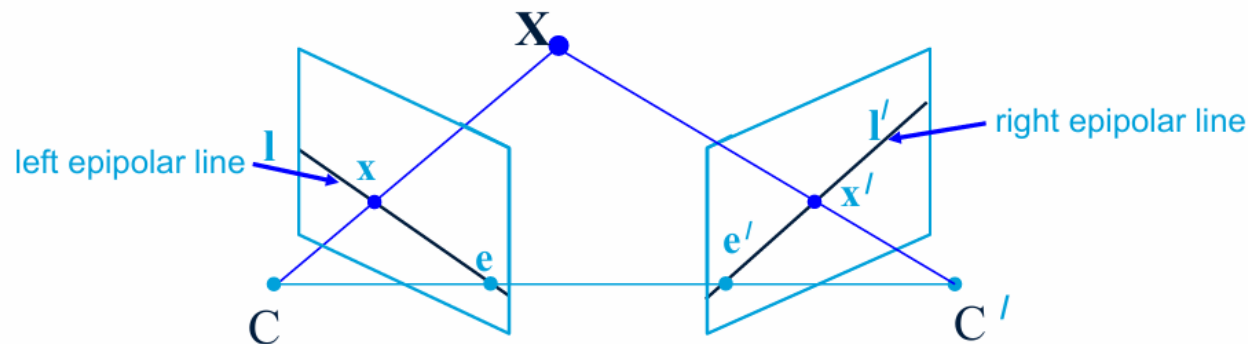
Epipolar Geometry

- The goal of **Epipolar Geometry** is to compute the 3D position of an object or set of points acquired by two distinct *calibrated* cameras with centers C, C'
$$x = PX \quad x' = P'X$$
- Epipolar Constraint:** a point in one camera generates an epipolar line on the other camera, the corresponding point lies somewhere on this line.
- The camera centers, the corresponding points x and x' , and the scene point X lie on a single plane, called the **Epipolar Plane**.



Epipolar Geometry

- The **Epipolar Line** l is the projection of the ray connecting X and C' and passes through x onto camera C .
- The **Epipole** e is the point of intersection of the line joining the camera centers (called the **Baseline**) with the image of C .
- In other words, an epipole is the projection of the center of the other camera:
$$e = PC' \quad e' = P'C$$
- Changing the position of X generates a family of epipolar planes, called the **Epipolar Pencil**.



Epipolar Geometry - Fundamental Matrix

- The epipolar geometry between two cameras can be described through the **Fundamental Matrix** F .
- $x'^T F x = 0$ (for corresponding image points)
- $l' = F x$ (for the epipolar line corresponding to x)
- $l = F^T x'$ (for the epipolar line corresponding to x')
- $F e = 0, F^T e' = 0$ (for the epipoles)

$$l = \begin{pmatrix} l_1 \\ l_2 \\ l_3 \end{pmatrix} \rightarrow l_1 x + l_2 y + l_3 = 0$$

Epipolar Geometry - Fundamental Matrix

- If you know the correspondences between two camera views, you can estimate the fundamental matrix using the **Eight-Point Algorithm**:

$$\begin{bmatrix} x & y & 1 \end{bmatrix} \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = 0 \Rightarrow f_{11}xx' + f_{12}xy' + \dots + f_{33} = 0$$

$$(x_1, y_1) \leftrightarrow (x'_1, y'_1)$$

$$(x_2, y_2) \leftrightarrow (x'_2, y'_2)$$

$$(x_3, y_3) \leftrightarrow (x'_3, y'_3)$$

$$\vdots$$

$$(x_n, y_n) \leftrightarrow (x'_n, y'_n)$$

$$\begin{bmatrix} x_1x'_1 & x_1y'_1 & x_1 & x'_1y_1 & y_1y'_1 & y_1 & x'_1 & y'_1 & 1 \\ x_2x'_2 & x_2y'_2 & x_2 & x'_2y_2 & y_2y'_2 & y_2 & x'_2 & y'_2 & 1 \\ x_3x'_3 & x_3y'_3 & x_3 & x'_3y_3 & y_3y'_3 & y_3 & x'_3 & y'_3 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_nx'_n & x_ny'_n & x_n & x'_ny_n & y_ny'_n & y_n & x'_n & y'_n & 1 \end{bmatrix} \begin{bmatrix} f_{11} \\ f_{12} \\ f_{13} \\ f_{21} \\ f_{22} \\ f_{23} \\ f_{31} \\ f_{32} \\ f_{33} \end{bmatrix} = 0$$

known

$Af = 0$

unknown

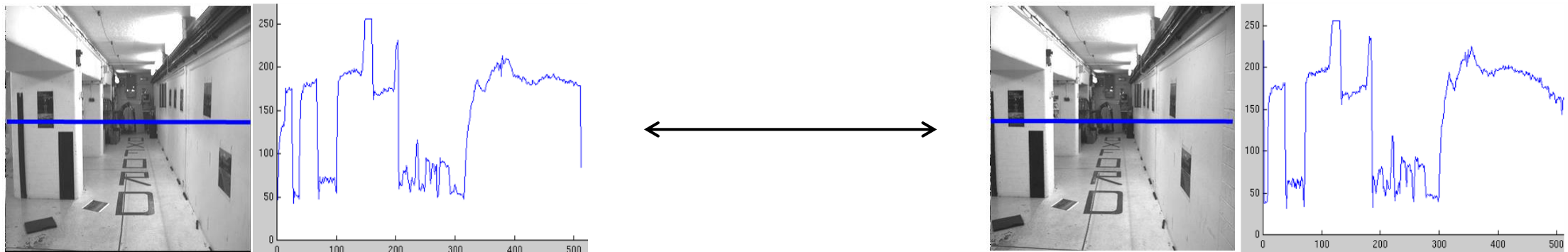
(At least 8 points are needed)

Epipolar Geometry - Fundamental Matrix

- Since finding an exact solution $Af = 0$ is unlikely given irreducible measurement noise, we instead minimize Af while enforcing $\|f\|^2 = 0$.
- **Lagrangian Cost:** minimize $\mathcal{L}(x) = x^T A^T A x - \lambda(x^T x - 1)$, where x is an eigenvector of $A^T A$ with eigenvalue $\lambda \rightarrow e_\lambda$.
- $\mathcal{L}(e_\lambda) = \lambda$, so we want the eigenvector x with the smallest eigenvalue.
- Use **Single Value Decomposition (SVD)** to find $A = UDV^T$, where the columns of V are the eigenvectors of $A^T A$ (the last one corresponds to the smallest eigenvalue).

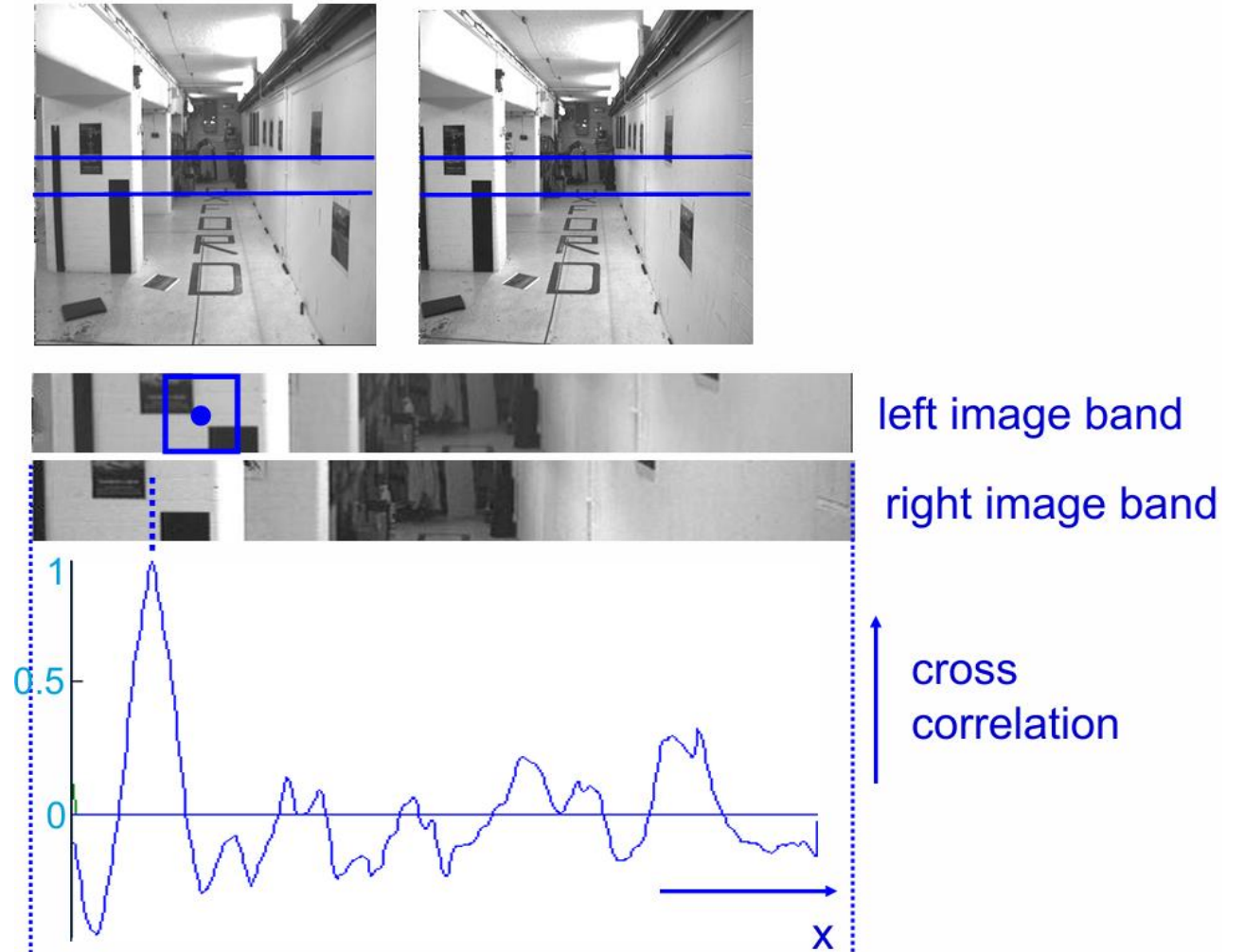
Stereo Correspondence

- Given two images and their respective cameras, how do you identify corresponding image points? This is the **Stereo Correspondence** problem.
- For each point in the left image, the corresponding point in the right image lies on the epipolar line $\rightarrow$ *1D search space*.
- **Photometric Constraint** (*assumption*): the intensity neighborhoods of corresponding points are similar across images $\rightarrow$ *measure similarity of neighborhood intensity using cross-correlation*.



Stereo Correspondence

1. Take a $k \times k$ neighborhood around the point as vector a and normalize (mean 0).
2. Compute cross-correlation (*cosine similarity*) between a and an equal patch on the other image along the epipolar line:

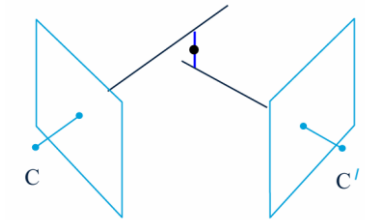


Stereo Correspondence - Triangulation

- In the presence of noise, back-projected rays do not intersect, and measured points do not lie on corresponding epipolar lines.

- Possible solutions:

1. Take the midpoint of the shortest line between the two rays.
2. **Linear Triangulation:** use $x = PX$ and $x' = P'X$ to solve for X . An exact solution in the form $AX = 0$ is likely not possible, so minimize AX (extends to more than two views).
3. **Lagrangian Cost:** minimize $\mathcal{L}(x) = x^T A^T A x - \lambda(x^T x - 1)$. x is an eigenvector of $A^T A$ with eigenvalue λ ($e_\lambda \rightarrow \mathcal{L}(e_\lambda) = \lambda \rightarrow$ we want x with the smallest eigenvalue. Use **Single Value Decomposition (SVD)** to find $A = UDV^T$, where the columns of V are the eigenvectors of $A^T A$ (the last one corresponds to the smallest eigenvalue).



Uncertainty increases as rays become more parallel. For nearby views feature matching becomes easier, while triangulation becomes harder.

